

EXPERIMENT 3

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Semester: 5

Subject Name: Computer Network

Subject Code: 24CSH-337

Experiment 3 – Study the workings of Hub, Switch, and Router.

Aim : To understand the working principles of Hub, Switch, and Router by creating and simulating network topologies in Cisco Packet Tracer and analyzing communication between connected devices

Objective

- To understand the functionality of a Hub, Switch, and Router and their roles in computer networking.
- To create and configure network topologies using Cisco Packet Tracer by connecting end devices through hubs, switches, and routers.
- To configure IP addressing and routing for enabling communication within the same network and between different networks.
- To analyze and compare data transmission through a Hub, Switch, and Router by observing packet flow and network behavior in simulation modeSoftware

Theory:

HUB:

1. Hubs works at the physical layer of OSI model
2. Used to set up LAN
3. Hub has multiple ports
4. Star Topology is developed with hub
5. When a packet arrives at one port, it is copied to the other ports so that all the segments of the LAN can see all packets.

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Switch:

1. A switch is a networking device that connects devices on a computer network to establish a local area network.
2. Unlike hub, Switch has memory. [It stores the MAC Address table into its memory].
3. Store MAC Address Table.
4. It is layer 2 Device for setting up the LAN.
5. It has n no of ports which are called as Interface / Ports. [Every port is given a number so that we can identify who is connected to which port with MAC Address].

Router:

1. A router is a networking device that forwards data packets between computer networks [between 2 or more LAN's] [2 different LAN's]
2. A router is connected to at least two networks, commonly two LANs or WANs or LAN and its ISP's Network.
3. It is a layer 3 device (Network Layer)
4. It also has memory which stores the routes in ROUTING TABLE.

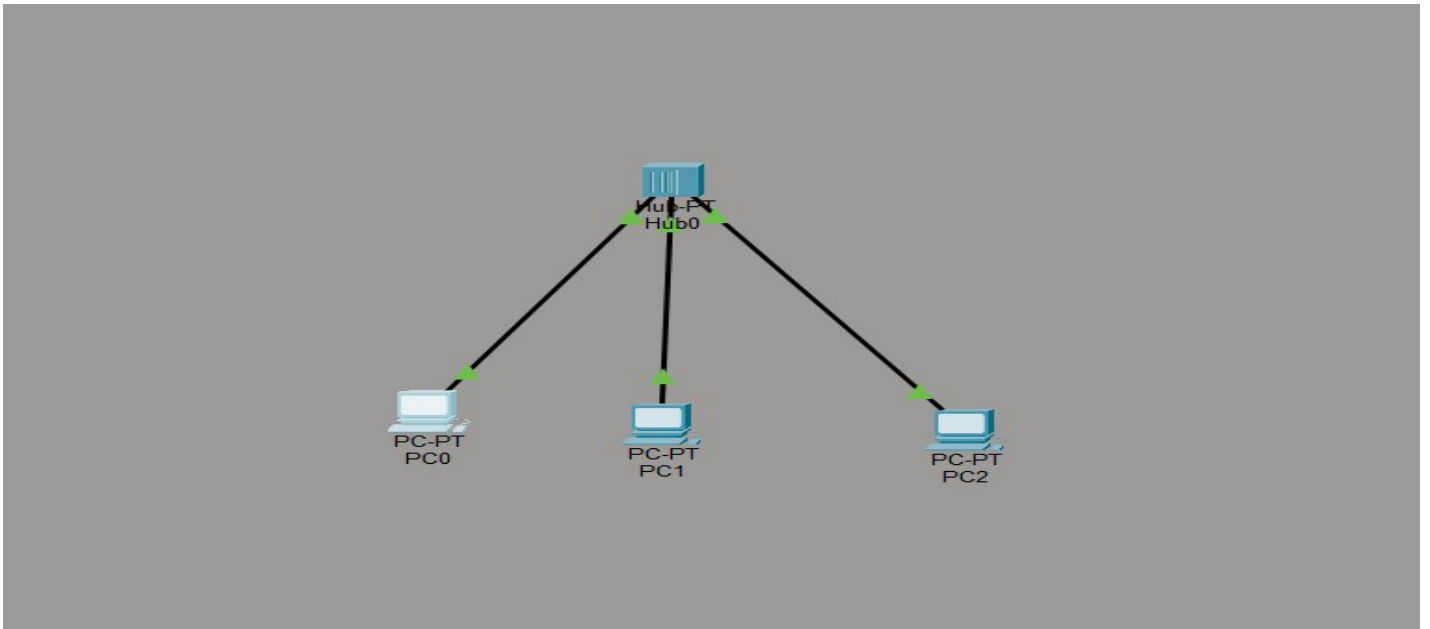
Practical / Experiment Steps

HUB:

STEPS:

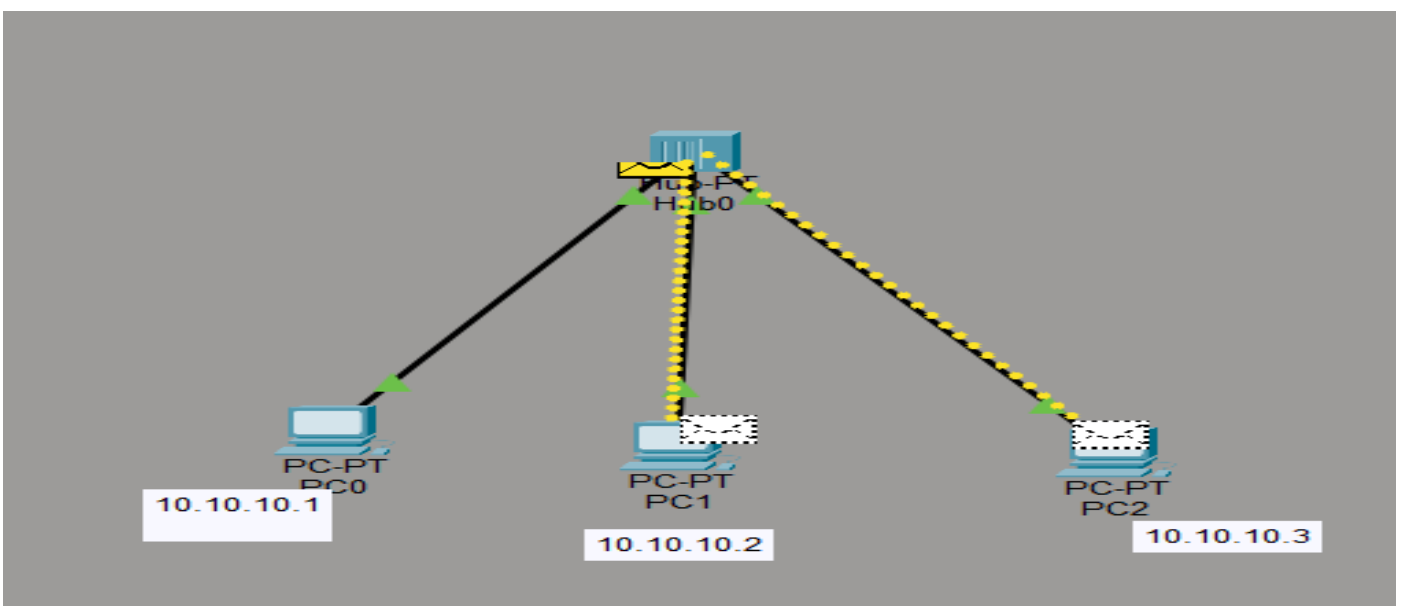
1. Hub section -> choose one hub -> click on hub and check for the ports
2. Take 3 PC's and take straight through cable -> Connect all PC's with Hub with FastEthernet port as shown in the pic 1.1

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[pic 1.1]

3. On bottom right corner click on simulation -> take one data packet -> take 10.10.10.1 as Source_ip & 10.10.10.3 as Destination_ip however the packet will be broadcasted to 10.10.10.2 as well as shown in the pic 1.2.



[pic 1.2]

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Advantage:

1. Cheaper than Switches.
2. Works good for smaller networks.

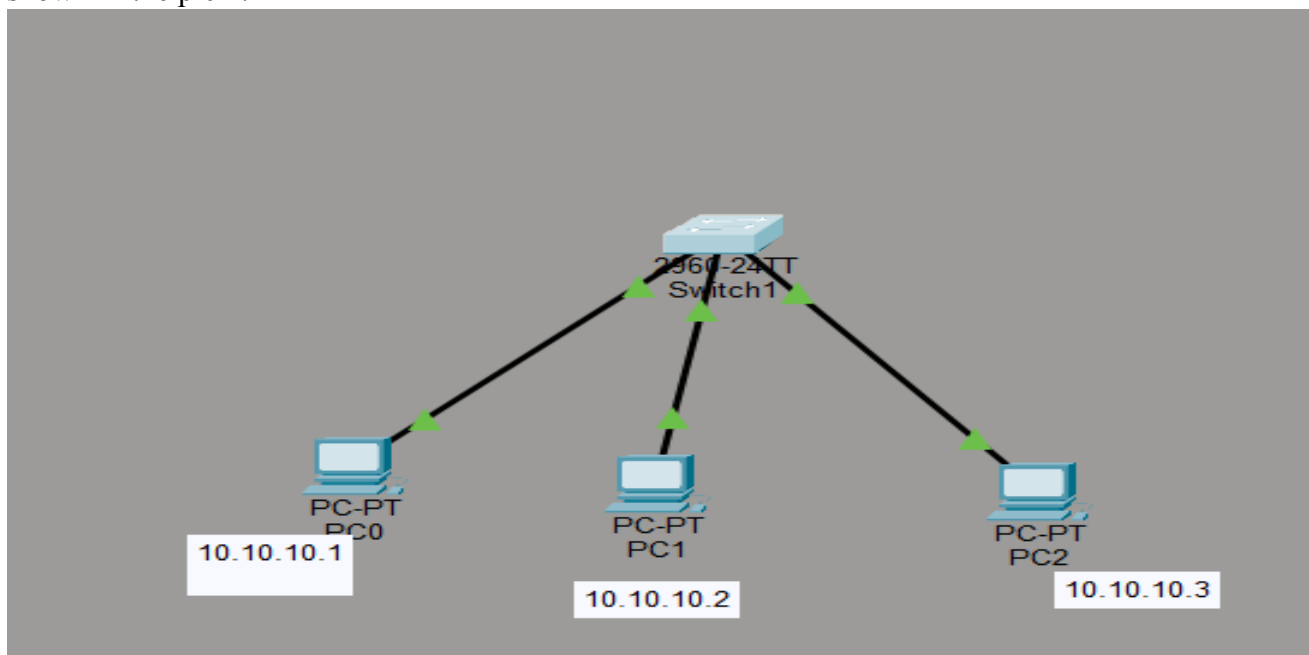
Disadvantages:

1. Issues with broadcasting.
2. No Memory.
3. Normally runs in half-duplex mode.

Switches:

Steps:

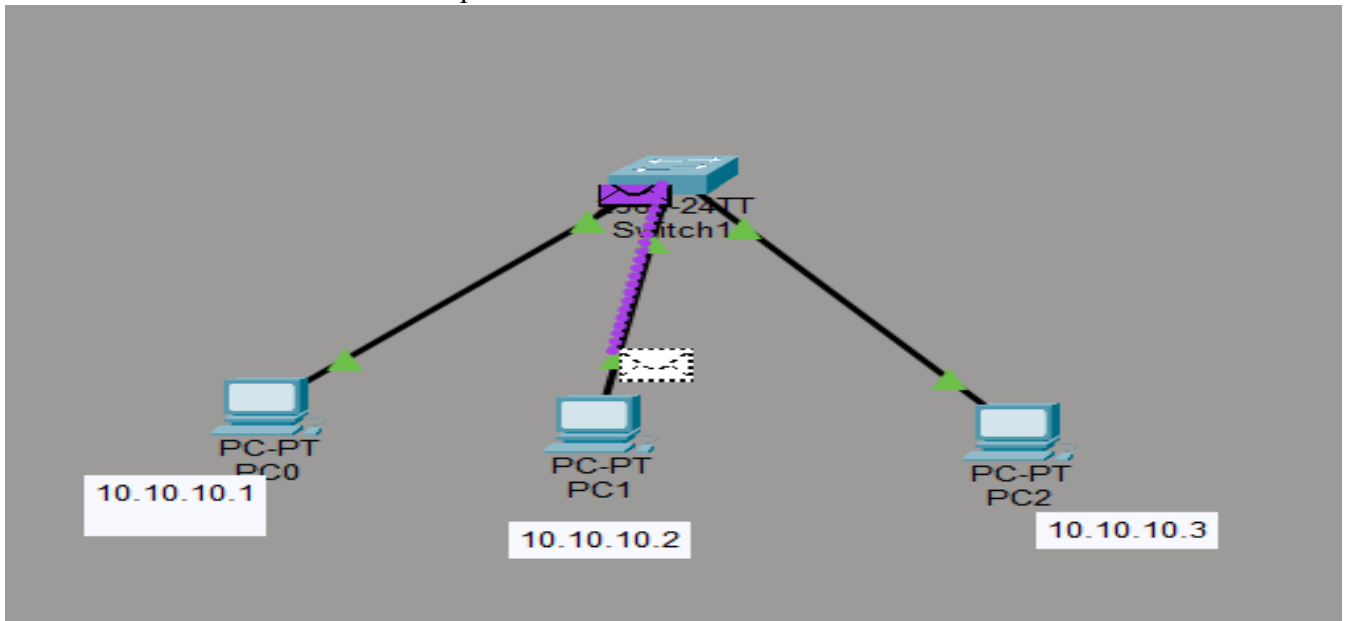
1. Switch section > choose one switch > click on switch and check for ports
2. Take 3 PC's and take straight through cable -> Connect all PC's with Switch with FastEthernet port as shown in the pic 2.1



[pic 2.1]

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3. On bottom right corner click on simulation -> take one data packet -> take 10.10.10.1 as Source_ip & 10.10.10.3 as Destination_ip and unlike hub it will not be broadcasted to all the other PCs and only be send to destination PC like shown in the pic 2.2



[pic 2.1]

Router:

Steps

1. Create two LANs using a Switch:

- LAN 1: IP 10.0.0.0 - Subnet Mask: 255.0.0.0
- Take 1 Switch & 3 PC's
- Configure the IPs of each PC as 10.0.0.1 for 1st PC, 10.0.0.2 for 2nd PC, 10.0.0.3 for 3rd PC
- Check this LAN 1 by sending a PDU from 10.0.0.1 to 10.0.0.3.

2. Create 2nd LANs using a Switch:

- LAN 2: IP 20.20.20.0 - Subnet Mask: 255.255.255.0
- Take 1 Switch & 3 PC's

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- Configure the IPs of each PC as 192.168.1.1 for 1st PC, 192.168.2 for 2nd PC, 192.168.3 for 3rd PC
- Check this LAN 2 by sending a PDU from 20.20.20.1 to 20.20.20.3.

3. Take one router (any router like Router 2911)

- Now this router has 3 interfaces: GigabitEthernet 0/0, GigabitEthernet 0/1, and GigabitEthernet 0/2 which means, we can connect three different LANs using these interfaces.
- Connect switch from LAN 1 with GigabitEthernet 0/1 with Router's GigabitEthernet 0/0.
- Connect switch from LAN 2 with GigabitEthernet 0/1 with Router's GigabitEthernet 0/1.

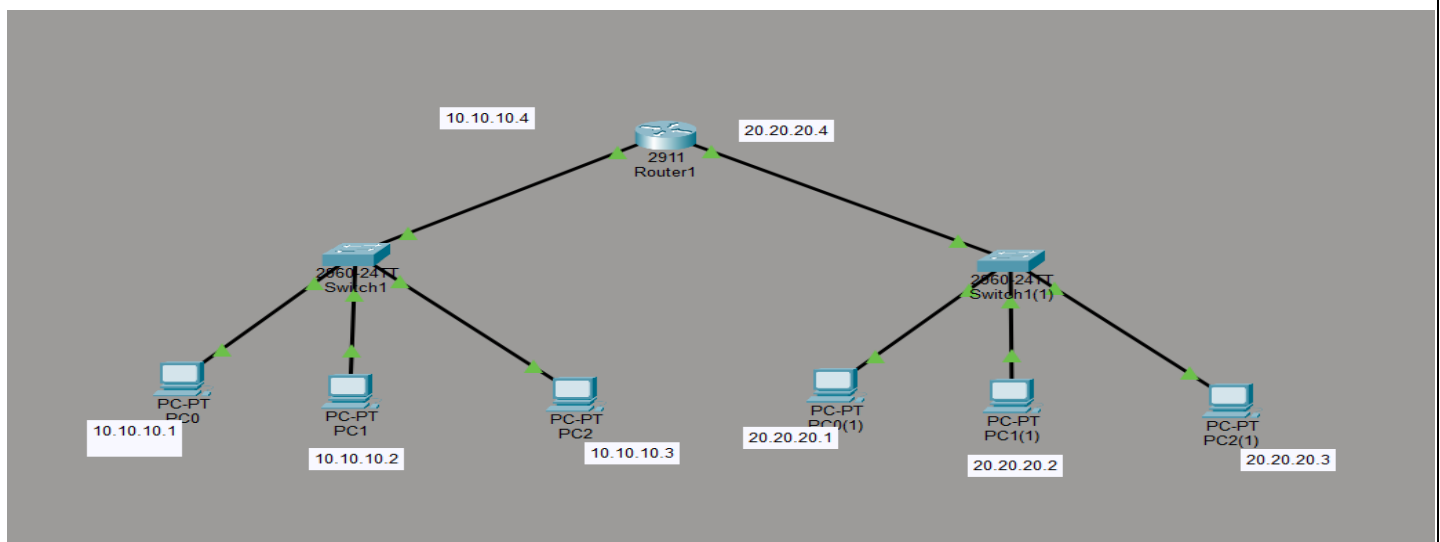
4. Now to send data from PC0 to PC5 we need to configure the default gateway as well for both LAN's.

-For LAN1: Gateway Address: 10.0.0.4

-For LAN2: Gateway Address: 20.20.20.4

- Now go to every PC of each LAN and configure the Gateway Address to each PC of each LAN.

[Click PC0 -> Go to Desktop -> IP Configuration -> Enter the Gateway Address in Default Gateway.



[pic 3.1]

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Learning Outcome

After completing this experiment, the learner will be able to:

1. **Identify and differentiate** the functions and characteristics of a Hub, Switch, and Router in a computer network.
2. **Design and simulate** basic network topologies using Cisco Packet Tracer by connecting end devices with networking devices.
3. **Configure IP addresses and routing settings** to establish communication within the same network and across different networks.
4. **Analyze packet transmission and network behavior** using Cisco Packet Tracer simulation mode and evaluate the role of each networking device in data communication.